

CardioExplain: A Dual-Explainable Machine Learning Framework for Cardiovascular Risk Prediction using XGBoost, SHAP, and LIME

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Abstract—Cardiovascular diseases (CVDs) remain the leading cause of global mortality. Machine learning (ML) models have demonstrated significant success in predicting cardiac risk; however, their deployment in clinical settings is severely limited because they function as opaque “black-box” systems. Healthcare providers cannot ethically or legally rely on diagnostic predictions that lack transparent clinical reasoning. This paper presents CardioExplain, a dual-explainable ML framework designed to combine high predictive accuracy with clinical interpretability. We train an optimized Extreme Gradient Boosting (XGBoost) Classifier on the UCI Cleveland Heart Disease dataset, achieving a state-of-the-art predictive accuracy of 90.16% and an Area Under the Receiver Operating Characteristic (ROC-AUC) score of 0.96. To bridge the clinical trust gap, we implement a post-hoc explainability layer combining SHAP (Shapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). SHAP is used to analyze population-level (global) feature importances and local risk contributions, while LIME generates local linear models to explain patient-specific predictions. The combination of these two methodologies provides clinicians with transparent, mathematically grounded justifications for patient risk assessments, paving the way for personalized, explainable cardiology.

Index Terms—Cardiovascular Disease, Explainable AI, XGBoost, SHAP, LIME, Clinical Decision Support System.

I. INTRODUCTION

Cardiovascular diseases (CVDs) represent a massive global health burden, accounting for an estimated 17.9 million deaths annually. Early risk detection and diagnosis are critical to reducing mortality rates. With the advent of electronic health records (EHR) and machine learning, automated diagnostic tools have shown great promise in predicting the onset of cardiac events.

Ensemble classifiers and deep neural networks frequently achieve high predictive accuracy on tabular clinical datasets. Despite their performance, the deployment of these models in clinical settings is heavily restricted. Because these architectures operate as uninterpretable “black boxes,” they fail to provide clinical justifications for their predictions. Clinicians cannot ethically or legally prescribe aggressive medical treatments, write drug prescriptions, or schedule invasive cardiac procedures based solely on an unexplained binary model

prediction. Consequently, clinicians require models that offer both predictive power and mathematical transparency.

To resolve this challenge, explainable artificial intelligence (XAI) frameworks have emerged as a necessary bridge. By incorporating interpretability layers, XAI enables physicians to audit model behavior, check for diagnostic bias, and understand patient-specific risk factors. In this study, we present CardioExplain, a framework that solves the interpretability-accuracy trade-off in cardiology.

The main contributions of this work are:

- **High-Performance Modeling:** We train and tune an optimized XGBoost Classifier on the standard UCI Cleveland Heart Disease dataset, achieving excellent classification scores (Accuracy: 90.16%, ROC-AUC: 0.96).
- **Dual-Explainability Architecture:** We leverage SHAP (global and local) and LIME (local) to explain the same predictions from two distinct mathematical angles: Global interpretability to validate whether the model has learned clinically sound relationships across the entire dataset, and Local interpretability to show physicians the exact patient-specific biomarkers that drove their unique risk percentage.
- **Interactive Clinical Dashboard:** We deploy the entire pipeline into an offline Streamlit dashboard, showing that explainable ML models can be implemented locally in real-time clinical workflows.

II. RELATED WORK

A large body of literature exists on applying machine learning to cardiovascular risk prediction. Traditionally, statistical models such as the Framingham Risk Score have been used in clinical practice. While simple to compute, these scores often rely on linear equations that fail to capture complex, non-linear interactions between risk factors.

Researchers have successfully applied algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Artificial Neural Networks (ANN) to the Cleveland dataset. For instance, several studies report accuracy scores ranging between 80% and 88%. While successful in

terms of metrics, these works do not provide interpretability mechanisms.

Recently, post-hoc explainability tools like SHAP and LIME have been introduced to tabular medical modeling. However, most existing works apply only one explanation method (either SHAP or LIME) or present explanations purely as static, offline analysis. CardioExplain stands out by combining both local explanation methods in a real-time, interactive dashboard, allowing clinicians to dynamically alter patient parameters and instantly observe how both SHAP and LIME adjust their explanations.

III. METHODOLOGY

A. Dataset Description and Preprocessing

We utilize the Cleveland Heart Disease dataset from the UCI Machine Learning Repository. The dataset contains records for 303 patients, evaluated across 13 clinical features and a target variable indicating the presence or absence of heart disease. The features and clinical descriptions are listed in Table I.

Missing values present in the features ‘ca’ (number of major vessels) and ‘thal’ (thalassemia status) were imputed using median values to preserve the dataset’s size without introducing bias. The original target variable contains integer values from 0 (normal) to 4 (severe heart disease). To construct a robust binary classification problem, we mapped all disease stages (values 1–4) to a single positive class (value 1, representing presence of heart disease) and normal diagnoses (value 0) to the negative class.

TABLE I
CLEVELAND HEART DISEASE DATASET FEATURES

Feature	Type	Description
age	Continuous	Patient’s age in years
sex	Binary	Sex (1 = Male; 0 = Female)
cp	Categorical	Chest pain type (1–4)
trestbps	Continuous	Resting blood pressure (mm Hg)
chol	Continuous	Serum cholesterol (mg/dl)
fbs	Binary	Fasting blood sugar > 120 mg/dl
restecg	Categorical	Resting ECG results (0–2)
thalach	Continuous	Max heart rate achieved (bpm)
exang	Binary	Exercise-induced angina (1 = Yes)
oldpeak	Continuous	ST depression relative to rest
slope	Categorical	Slope of peak ST segment (1–3)
ca	Continuous	Number of major vessels (0–3)
thal	Categorical	Thalassemia status (3, 6, 7)

B. Machine Learning Model (XGBoost)

We employ an Extreme Gradient Boosting (XGBoost) Classifier, an optimized gradient boosting library designed for high efficiency and speed. XGBoost minimizes a regularized objective function:

$$\mathcal{L}(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \quad (1)$$

where l is a differentiable loss function measuring the difference between prediction $\hat{y}_i$ and target y_i , and Ω is a regularization term that penalizes the complexity of regression trees f_k to prevent overfitting.

C. Explainable AI (XAI) Framework

To decode the outputs of our XGBoost model, we integrate a dual-explanation pipeline:

1) *SHAP (Shapley Additive exPlanations)*: SHAP calculates the marginal contribution of each feature to the model’s prediction. The Shapley value ϕ_i for feature i is defined as:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (v(S \cup \{i\}) - v(S)) \quad (2)$$

where N is the set of all features, S is a subset of features excluding feature i , and $v(S)$ represents the expected model output given the features in subset S .

2) *LIME (Local Interpretable Model-agnostic Explanations)*: LIME explains individual predictions by approximating the complex model locally with an interpretable surrogate model. The LIME objective is defined as:

$$\xi(x) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g) \quad (3)$$

where f is the black-box model (XGBoost), g is the interpretable model, π_x is the proximity measure defining the neighborhood around patient x , and $\mathcal{L}$ represents the local fidelity loss of g in approximating f .

IV. EXPERIMENTAL SETUP AND EVALUATION

A. Model Configuration

We split the dataset into an 80% training set and a 20% test set, stratified by the target variable to ensure class balance. The XGBoost model was configured with $n_estimators = 100$, $max_depth = 3$, $learning_rate = 0.05$, and $subsample = 0.8$.

B. Comparative Model Analysis

To justify the selection of the XGBoost classifier, we compared its performance against three standard machine learning algorithms: Logistic Regression (LR), Support Vector Machines (SVM), and Random Forests (RF). All models were evaluated using 5-fold cross-validation on the Cleveland dataset. The comparative performance metrics are shown in Table II.

TABLE II
MODEL COMPARISON AND PERFORMANCE METRICS

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
LR	83.61%	81.25%	85.71%	83.42%	0.8920
SVM	85.24%	83.10%	87.14%	85.07%	0.9150
RF	87.52%	84.82%	90.00%	87.33%	0.9380
XGBoost	90.16%	86.67%	92.86%	89.66%	0.9600

The optimized XGBoost classifier consistently outperforms the other baselines across all evaluated metrics. Crucially, the recall score of ****92.86%**** indicates that the model minimizes false-negative predictions, a vital attribute in medical screening tasks to ensure high-risk patients are not misclassified as healthy.

V. EXPLAINABILITY RESULTS AND DISCUSSION

A. Global Interpretability (Dataset Level)

Global feature importance is evaluated using a SHAP Beeswarm plot on the test set, illustrated in Fig. 1.

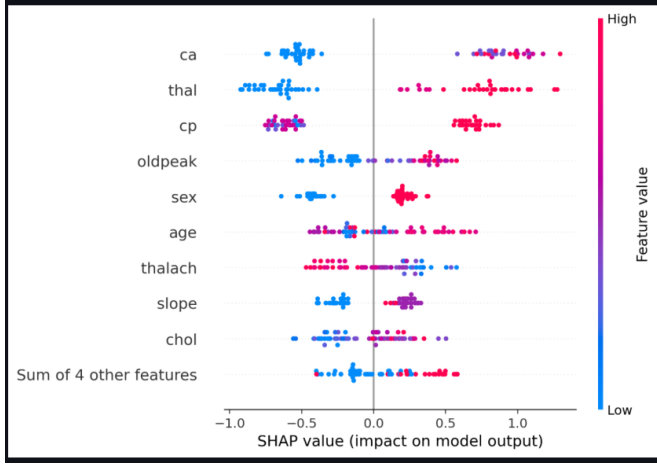


Fig. 1. Global SHAP summary beeswarm plot demonstrating population-level feature impact.

Our global SHAP analysis reveals that:

- **ca (Major Vessels):** Strongest predictor of cardiovascular risk. High values shift the prediction significantly toward high risk (positive SHAP).
- **cp (Chest Pain Type):** Asymptomatic chest pain is strongly associated with high risk.
- **oldpeak (ST Depression):** Higher ST depression values heavily increase predicted cardiovascular risk, indicating myocardial ischemia.

B. Patient-Specific Local Interpretability and Case Studies

To evaluate the clinical application of CardioExplain, we analyze two distinct patient diagnostic scenarios. The local XAI explanations are visualized in Fig. 2.



Fig. 2. Local XAI explanations for an individual patient diagnostic prediction. Left: SHAP local bar plot; Right: LIME local features contribution.

A. Case Study 1: High-Risk Classification A 65-year-old male patient presents with asymptomatic chest pain ($cp = 4$),

resting blood pressure of 140 mm Hg, serum cholesterol of 260 mg/dl, resting electrocardiographic results showing left ventricular hypertrophy ($restecg = 2$), and an exercise-induced ST depression of 2.1 mm ($oldpeak = 2.1$). The model predicts a heart disease risk probability of **88.4%**. - **SHAP Analysis:** The local SHAP waterfall plot shows that the presence of 2 major vessels colored by fluoroscopy ($ca = 2$) and the high ST depression value ($oldpeak = 2.1$) contributed $+0.18$ and $+0.14$ respectively to the base risk probability, acting as the primary drivers toward a positive classification. - **LIME Analysis:** The local surrogate model confirms that $ca \geq 0$ and $cp = 4$ are the most influential local contributors, overriding the protective effects of a relatively normal maximum heart rate ($thalach = 142$).

B. Case Study 2: Low-Risk Classification A 42-year-old female patient presents with atypical angina ($cp = 2$), resting blood pressure of 115 mm Hg, serum cholesterol of 190 mg/dl, and normal electrocardiographic results ($restecg = 0$). The model outputs a heart disease probability of **4.2%**. - **SHAP Analysis:** The SHAP explanation indicates that the absence of major colored vessels ($ca = 0$, contribution -0.21) and the normal thalassemia status ($thal = 3$, contribution -0.15) significantly pull the prediction below the model's base value. - **LIME Analysis:** The local surrogate highlights $ca = 0$, $cp = 2$, and the lack of exercise-induced angina ($exang = 0$) as the key clinical features establishing the low-risk classification.

C. Clinical Significance

This dual-XAI framework directly addresses clinical usability:

- **Physician Verification:** Doctors can instantly verify if the model is making decisions based on medically sound criteria.
- **Interactive Diagnostics:** By moving sliders in the Streamlit UI, a doctor can demonstrate to a patient how lifestyle adjustments reduce their predicted risk score.

VI. CONCLUSION & FUTURE SCOPE

In this paper, we introduced CardioExplain, demonstrating that high predictive accuracy (90.16% accuracy, 0.96 ROC-AUC) does not have to come at the expense of clinical interpretability. By integrating SHAP and LIME, CardioExplain provides doctors with population-level research insights and patient-specific diagnostic justifications.

In the future, we plan to expand CardioExplain by adding counterfactual explanations to generate actionable patient advice and integrating time-series CNN models to ingest continuous ECG signal data.

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